

On integral rational representation of the invariants of a system of arbitrarily many basic forms

At the end of his contribution to the Schwarz Festschrift, *On the invariants of a system of arbitrarily many basic forms*, Hilbert expressed the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ) , where J denotes a complete system of invariants for the linearly independent basic forms and PJ denotes the invariants obtained from J by polar processes.

In what follows I show, in Part I, that this conjecture is indeed correct, and in fact as a simple consequence of the reduction theorem: every form in arbitrarily many rows of n variables can be represented as a sum of polars of forms that involve only n fixed rows.

In Part II I give a more conceptual proof of the reduction theorem by interpreting the question as a problem of equivalence of linear families of forms. I owe this point of view, and likewise an essential part of the reasoning employed, to Fischer's paper *On algebraic systems of modules* and to unpublished propositions of E. Fischer, whose use he kindly permitted me. Finally, in Part III I indicate how analogous considerations also lead to the most general series expansions.

I. The invariant-theoretic consequence of the reduction theorem

Let Θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \dots, A_N,$$

where in general the row A_i consists of the n elements

$$a_1^{(i)}, \dots, a_n^{(i)}.$$

The coefficients of Θ may be indeterminates or quantities belonging to a given field of rationality. Let P denote an integral rational function—with rational numerical coefficients—of the polar operators

$$P_{ih} = a_1^{(i)} \frac{\partial}{\partial a_1^{(h)}} + a_2^{(i)} \frac{\partial}{\partial a_2^{(h)}} + \dots + a_n^{(i)} \frac{\partial}{\partial a_n^{(h)}},$$

where by a product such as $P_{ih}P_{jk}\Theta$ one understands the action of P_{ih} on $P_{jk}\Theta$.

Then the reduction theorem referred to above is expressed by the identity

$$\Theta = \sum P\Theta_0, \tag{1}$$

where the forms Θ_0 involve only the rows

$$A_1, A_2, \dots, A_n$$

and are obtained from Θ by polar processes P .

Now consider the basic system (F) consisting of the N forms of the same order and the same number of variables,

$$F_1, F_2, \dots, F_N,$$

whose coefficients are taken respectively to be the N rows

$$A_1, A_2, \dots, A_N.$$

Let (J) denote a complete system of invariants of

$$F_1, \dots, F_n,$$

that is, a finite set of invariants such that every invariant of

$$F_1, \dots, F_n$$

can be expressed integrally and rationally through the invariants (J) . Hilbert's conjecture to be proved is then the following: for the simultaneous invariants S of

$$F_1, \dots, F_N$$

a complete system is given by the invariants (J, PJ) .

Applying the reduction theorem (1) to a simultaneous invariant S —which can itself be regarded as a form in the N rows A_i —one sees at once that every simultaneous invariant is representable as a sum of polars of invariants that involve only the n rows

$$A_1, \dots, A_n.$$

But such forms are invariants of

$$F_1, \dots, F_n,$$

and hence can be expressed integrally and rationally through the invariants (J) . On the other hand, by the very definition of the polar processes, every form obtained from an invariant of

$$F_1, \dots, F_n$$

by a polar process is again a simultaneous invariant of

$$F_1, \dots, F_N.$$

Since the polar process preserves integral rational expressibility, it follows that every simultaneous

invariant S of

$$F_1, \dots, F_N$$

can be represented integrally and rationally through the invariants (J, PJ) .

Thus we have proved:

Theorem. Every simultaneous invariant of a system of arbitrarily many basic forms is integrally and rationally expressible through the finite family (J, PJ) , where J is a complete invariant system for n linearly independent basic forms and PJ denotes all invariants obtained from J by polar operations.

This establishes Hilbert's conjecture.

In the next batch I will continue with Noether's more conceptual proof of the reduction theorem via equivalence of linear families of forms.